Oil Sales Analysis — Executive Summary

Introduction
Approach: I prepared a concise summary and built a simple predictive model to estimate monthly value sales. Data: `data/cleaned_data.csv` (aggregated sales by store/sku/month). Assumptions: data rows represent monthly store-SKU observations; modelling uses month, size, price and top brands as predictors.

Methods
Cleaning steps: numeric coercion for sales/price/size, dropped rows missing `value_sales`, constructed a date from year/month where present. Analysis: computed aggregates (total sales, top cities/brands, monthly seasonality). Model selection: a RandomForestRegressor was used as a robust baseline (handles nonlinearity and mixed features). Evaluation: random 80/20 train-test split; reported R^2 and RMSE on the holdout set.

Findings
Records analysed: 2,000
Total value sales: 1,229,678.71
Total volume sold: 19,944.2
Top cities by value sales:
- AL BAHA: 143,558.12
- MAKKAH: 116,187.24
- DAMMAM: 104,942.31
- TABUK: 104,204.55
- YANBU: 101,609.15

Top brands by value sales:
- GULF GOLD: 144,031.09
- SABAYA: 135,717.25
- ZAHRA: 134,058.99
- RIMAL: 130,336.90
- NAJMA: 124,916.64

Observed seasonality with peak aggregate sales in month: 12
Model training/evaluation was not completed due to insufficient/invalid features.

Top 5 Cities — value sales

City	Value Sales
AL BAHA	143,558.12
MAKKAH	116,187.24
DAMMAM	104,942.31
TABUK	104,204.55
YANBU	101,609.15

Recommendations
1) Use the baseline model for short-term forecasting and re-train monthly with new data; include promotional flags and price changes to improve accuracy. 2) Prioritise inventory and promotions in the top cities/brands identified — these drive a large share of value sales. 3) Investigate observed seasonality (peak month) and align stock and marketing spend accordingly. 4) For production: add time-series features (lagged sales), store-level historical trends, and external signals (holidays, oil price) to move from a baseline model to a higher-performing model.